

Jeffrey Xing

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Summary

Researcher interested broadly in the intersection between vocal communication, emotion perception, and sound preference. I take a highly interdisciplinary approach to my research, which ranges from modeling basic perceptual processes using machine-learning, testing animals/humans in natural vocal communication tasks using qualitative/quantitative methods, scanning/analyzing brain-imaging data, to assessing/informing medical student empathy training.

Education

University of California San Diego

Ph.D. in Experimental Psychology Oct 2022 – Present

M.S. in Experimental Psychology Oct 2022 – July 2025

- Coursework: Quantitative Methods in Psychology, Sensation & Perception, Biological Psychology, Methods of Human Cognitive Neuroscience, Psycholinguistics, Principles of Behavior Evolution, Computational Psychology, Magnetoencephalography

B.S. in Psychology with a Specialization in Sensation and Perception Oct 2018 – June 2022

- *Summa cum laude*, GPA: 4.0

B.A. in Interdisciplinary Computing and the Arts Major (Music) Oct 2018 – June 2022

Experience

Graduate Student Researcher, Gentner Lab – La Jolla, CA Oct 2022 – Present

- Developed mixed method behavioral experiments that generated insights on how people continuously track in real-time other people's emotion through speech
- Designed neuroimaging experiments for studying neurobiology of speech perception with neuroscience and linguistic collaborators
- Currently designing mixed methods studies on how people perceive physician empathy in real-time to inform medical student empathy training
- Modeled basic auditory perception using machine-learning techniques and conducted studies on how acoustic predictability relates to sound preference

Undergraduate Research Assistant, Gentner Lab – La Jolla, CA Aug 2020 – Jun 2022

- Annotated birdsong acoustics and analyzed them using information theoretic and rhythmic approaches

Recording Assistant, Department of Music, UCSD – La Jolla, CA Oct 2019 – Jun 2022

- Produced and archived concert recordings for contemporary music research

Stage Crew, Department of Music, UCSD – La Jolla, CA Nov 2018 – Jun 2022

- Assisted technical audio setups for concerts

Publications

Xing, J., Sainburg, T., Taylor, H., & Gentner, T. Q. (2022). Syntactic modulation of rhythm in Australian pied butcherbird song. *Royal Society Open Science*, 9(9), 220704. <https://doi.org/10.1098/rsos.220704>

Conference Papers

Xing, J., Sainburg, T., Gentner, T. Q., & Taylor, H. (2021, February). Examining song complexity of Australian pied butcherbirds. In *Proceedings of Acoustics* (Vol. 21, No. 23, p. 2022).

Presentations

Xing, J., Ostrowski, L.M., Lang, B., Urrutia, K., Huang, M., & Gentner, T. Q. (2025) Cortical temporal dynamics of speech perception: Dissociating linguistic and paralinguistic elements. Poster Presentation. Society for Neuroscience (November 2025)

Lang, B., **Xing, J.,** Ostrowski, L.M., Urrutia, K., Huang, M., & Gentner, T. Q. (2025) Cortical temporal dynamics of tracking linguistic and paralinguistic elements. Poster Presentation. Society for Neurobiology of Language (September 2025)

Fellowship and Awards

National Science Foundation Graduate Research Fellow (NSF GRFP)	2024 - Present
• Fully-funded graduate fellowship in recognition for potential in impactful STEM research	
Norman Henry Anderson Fellowship , Department of Psychology, UCSD	2022 - 2023
Norman Henry Anderson Honors Thesis Award , Department of Psychology, UCSD	2022
Distinguished Undergraduate Scholar , Department of Psychology, UCSD	2021 - 2022
Psychology Honors Program , Department of Psychology, UCSD	2021 - 2022
Richard L. and Fern W. Erion and Laidlaw-Erion Scholarship , UCSD	2020 - 2022
William Stout Scholarship , Department of Psychology, UCSD	2021 - 2022
Academic Enrichment Program Summer Research Scholarship , UCSD	2021
Research Perseverance During COVID Award , Department of Psychology, UCSD	2021
Halicioğlu Data Science Institute Undergraduate Scholarship , Halicioğlu Data Science Institute, UCSD	2021
Diana C. Miles Scholarship , UCSD	2020 - 2021
UC San Diego Provost Honors , UCSD	2018 - 2022

Academic Service

Colloquium Representative , Department of Psychology, UCSD	2023-2025
• Organized schedules and hospitality for 7 visiting researchers	
Faculty Recruitment Representative , Department of Psychology, UCSD	2023-2024
• Solicited and represented graduate student feedback in 4 UCSD Psychology hiring decisions	

Teaching and Mentorship

Teaching Assistant , Department of Psychology, UCSD	2023-2024
• Psychology of Human Sexuality, Memory and Amnesia, Social Cognition, Impulse Control Disorders	
Teaching Workshop Series , Department of Psychology, UCSD	2023
• Completed workshop on evidence-based teaching strategies, assessment and student support	
Undergraduate Instructional Apprentice , Department of Psychology, UCSD	2021 - 2022
• Research Methods I & II, Introduction to Statistics	
CoB-KIBM Scholars Program Mentor , Colors of the Brain, Kavli Institute for Brain and Mind, UCSD	2025
• Helped select and directly mentored 2 CoB-KIBM Scholars for summer 2025	
Undergraduate Research Mentor , Department of Psychology, UCSD	2022 - Present
• Maya Langer, Haverford College, Undergraduate in Biology	
• Chioma Amadi, UC San Diego, Undergraduate in Cognitive & Behavioral Neuroscience	
• David Wang, UC San Diego, Undergraduate in Computer Engineering	

- Annabelle Lee, UC San Diego, Undergraduate in Biochemistry
- Amna Sharif, UC San Diego, Undergraduate in Cognitive & Behavioral Neuroscience (Graduated)
- Kate Urrutia, UCSD Psychology Independent Research Excellence Award Recipient (2025), UC San Diego, Undergraduate in Cognitive & Behavioral Neuroscience
- Mohini Kanumuri, UC San Diego, Undergraduate in Clinical Psychology
- Jack Celaya, UC San Diego, Undergraduate in Cognitive & Behavioral Neuroscience
- Cat Brown, UC San Diego, Undergraduate in Linguistics
- Giselle Lizaola, UC San Diego, Undergraduate in Linguistics

Skills

Quantitative Research

- Psychoacoustics, Behavioral Testing, Psychometrics

Neuroscience

- Neuroimaging, Magnetoencephalography (MEG), Magnetic Resonance Imaging (MRI), Computational Neuroscience, Animal Behavior

Qualitative Research

- Surveys, Open-response Questions, Video Observations, Focus group (In-progress)

Programming

- Linux, Python, R

Statistics

- General Linear Modeling, Nonparametric Statistics, Time-series Analyses, Reliability Analyses, Machine-Learning

Audio Engineering

- Psychoacoustics, Digital Audio Workstations, Contemporary Music Composition, Music Production, Sound Synthesis, Sound Effect, Recording Studio Setup, Live-Mixing, Mixing/Mastering, 3D/Binaural Audio

Hardware Prototyping

- Arduino, RaspberryPi, Autodesk Fusion 360, Circuit Board Assembly, 3D Printing

Science Communication

- Data Visualization, Contextualizing Pain-Point, Crafting Perceiver Narrative, Audience-specific presentation framing, Scientific Collaborations

Languages

- English, Chinese

Interests

Auditory Perception

- Predictive Coding, Habituation and Stimulus Recovery, Rhythmic Expressivity, Sound Complexity, Auditory Illusions, Spatial Audio

Emotion Perception

- Real-time Speech Emotion, Emotional Prosody, Emotional Semantics, Physician Empathy and Compassion

Behavioral Preference

- Sound Preference and Attraction, Incentive Salience

Neurobiology of Language

- Paralinguistic Perception, Linguistic Abstraction, Dynamical Systems Modeling

Machine Learning

- State Space Sequence Modeling, Speech Emotion Recognition, Topic/Semantic Modeling